

Hunting a Counterexample to Agrawal’s Conjecture: Background and Strategy

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1 The conjecture and its status

For n, r coprime, write $T(-1, n, r)$ for the congruence

$$(X - 1)^n \equiv X^n - 1 \pmod{n, X^r - 1}. \quad (1)$$

Conjecture 1 (Agrawal; Bhattacharjee–Pandey 2001). *If r is a prime not dividing n and (1) holds, then n is prime or $n^2 \equiv 1 \pmod{r}$.*

If true, a single congruence (1) at one $r = O(\log n)$ certifies primality, giving a deterministic $\tilde{O}(\log^3 n)$ test — versus $\tilde{O}(\log^6 n)$ for the best proven variant (Lenstra–Pomerance Gaussian periods [10]). The status is peculiar: Lenstra and Pomerance [2] gave a heuristic predicting $\gg x^{1-\varepsilon}$ counterexamples below x , yet none is known; exhaustive search (Primaboinca, 2010–2020) cleared $n < 10^{17}$, and folklore places the smallest counterexample at hundreds of digits. Popovych [3] proposed the repair of adding $(X + 2)^n \equiv X^n + 2$ as a second hypothesis; that variant is untouched by any heuristic or search.

Our angle: abandon the smallest counterexample. Any counterexample disproves the conjecture, the heuristic density improves with size, and the construction below is deterministic — if its arithmetic side conditions hold, the congruence (1) is guaranteed.

2 The Lenstra–Pomerance construction ($r = 5$)

Theorem 1 (Lenstra–Pomerance [2]; statement as in Váňa [4], who supplies the $k \equiv 3$ case). *Let $n = p_1 \cdots p_k$ with p_i distinct primes such that*

- (a) k is odd;
- (b) $p_i \equiv 3 \pmod{80}$ for all i ;
- (c) $p_i - 1 \mid n - 1$ for all i (n is a Carmichael number);
- (d) $p_i + 1 \mid n + 1$ for all i (n is a Lucas–Carmichael number).

Then $T(-1, n, 5)$ holds and $n^2 \not\equiv 1 \pmod{5}$.

Why this works, in one paragraph. Since $p \equiv 3 \pmod{5}$ has order 4 in $(\mathbb{Z}/5)^\times$, over $\mathbb{F}_p$ we have $X^5 - 1 = (X - 1)\Phi_5(X)$ with Φ_5 irreducible, so $\mathbb{Z}[X]/(p, X^5 - 1) \cong \mathbb{F}_p \times \mathbb{F}_{p^4}$. The $\mathbb{F}_p$ component of (1) is trivial. In $\mathbb{F}_{p^4}$, with ζ a primitive fifth root of unity, Frobenius gives $(\zeta - 1)^{p^j} = \zeta^{p^j} - 1$ for all j . Conditions (c),(d) say $n \equiv 1 \pmod{p - 1}$ and $n \equiv -1 \pmod{p + 1}$, i.e. $n \equiv p^3 \pmod{\text{lcm}(p - 1, p + 1)}$; the mod-80 condition pins the 2-adic ambiguity, and one checks (Váňa’s $\rho(p) \mid 10(p^2 - 1)$, [4, Thm. 3.4]) that control modulo $10(p^2 - 1)$ — not $p^4 - 1$ — already forces $(\zeta - 1)^n = \zeta^n - 1$. Finally $p_i \equiv 3 \pmod{5}$ and k odd give $n \equiv 3^k \in \{2, 3\} \pmod{5}$, so $n^2 \not\equiv 1 \pmod{5}$.

The pain is concentrated in (c)∧(d):

Lemma 1 (Side-coprimality). *If n satisfies (c) and (d), then $\gcd(p_i - 1, p_j + 1) \mid 2$ for all i, j .*

Proof. An odd prime q dividing both divides $n - 1$ and $n + 1$, hence 2. □

So the odd primes supporting all the $p_i - 1$ and those supporting all the $p_i + 1$ must be *disjoint sets*. No number that is simultaneously Carmichael and Lucas–Carmichael is known (none exists below 10^{18} , by Pinch’s tables [7]); Lemma 1 is the structural reason, and the reason the smallest example is expected to be enormous. Note (b) gives $v_2(p - 1) = 1$, $v_2(p + 1) = 2$, $5 \nmid p^2 - 1$, and for k odd the 2-adic parts of (c),(d) are automatic ($n \equiv 3 \pmod{4}$); the problem lives entirely in odd moduli prime to 5.

3 Reduction to a subset-product problem

This is now a job for the Erdős-style machinery behind large-scale Carmichael constructions [5, 6]. Fix disjoint sets Q_-, Q_+ of odd primes $\neq 5$ and harvest the pool

$$\mathcal{P} = \{p \leq x : p \equiv 3 \pmod{80}, \frac{p-1}{2} \text{ is } Q_- \text{-smooth}, \frac{p+1}{4} \text{ is } Q_+ \text{-smooth}\}.$$

Put $\Lambda_- = 2 \operatorname{lcm}\{p - 1\}$, $\Lambda_+ = 4 \operatorname{lcm}\{p + 1\}$ over $\mathcal{P}$. For any $S \subseteq \mathcal{P}$ with $|S|$ odd, $|S| \geq 3$, the number $n = \prod_{p \in S} p$ satisfies Theorem 1 provided

$$n \equiv 1 \pmod{\Lambda_-} \quad \text{and} \quad n \equiv -1 \pmod{\Lambda_+}, \tag{2}$$

a condition *stronger* than (c),(d) but linear in the exponent vector: taking discrete logarithms in each cyclic component of $(\mathbb{Z}/\Lambda_-)^\times \times (\mathbb{Z}/\Lambda_+)^\times$ (Pohlig–Hellman is cheap — all orders are smooth by design), (2) becomes a 0/1 subset-sum over $\bigoplus_i \mathbb{Z}/m_i$ with one extra $\mathbb{Z}/2$ component for the parity of $|S|$. Heuristically solvable once

$$2^{|\mathcal{P}|} \gg |G|, \quad |G| \approx \varphi(\Lambda_-) \varphi(\Lambda_+),$$

i.e. $|\mathcal{P}| \gtrsim \log_2 \Lambda_- + \log_2 \Lambda_+$ with slack for the solver — ballpark: a 15–30k-bit group wants a pool of 50–100k primes. Solvers, in escalating order: staged CRT à la Löh–Niebuhr [6], Wagner’s k -list generalized birthday algorithm [8] (the pool-rich regime is exactly its habitat), lattice reduction only for terminal low-dimensional corrections.

Two design points. *Harvesting is constructive, not sieved*: enumerate Q_- -smooth m , set $p = 2m + 1$, test $p \equiv 3 \pmod{80}$ and primality, then trial-divide $(p + 1)/4$ over Q_+ ; this reaches $x = 10^{12+}$ easily. An *asymmetric split* looks best: $Q_+ = \{\text{odd primes} \leq B\} \setminus \{5\}$ (the “random side”, where $(p + 1)/4$ -smoothness is Dickman-governed), $Q_- \subset (B, B']$ medium primes (the “constructive side”, disjointness automatic). The resulting n would have thousands of prime factors and 10^4 – 10^5 digits; verifying (1) directly at that size is a few core-hours, and testing Popovych’s second congruence on any find is free — either we kill both conjectures or we get the first data separating them.

4 Odds, honestly

For: the double-condition pipeline appears never to have been run at scale — Vána mined existing Carmichael tables (wrong haystack), Hegde–Devaraj [9] stopped at density heuristics, and the modern subset-sum toolkit postdates the 2003 construction. Against: Lemma 1 thins the pool — both-sided smoothness costs roughly $\rho(u)^2$ (Dickman) *before* the disjointness tax, and if $\log |G|$ grows linearly with the pool the feasibility inequality never closes. Phase 1 of PLAN.md measures exactly this and produces a go/no-go number before any solver is built; a quantified infeasibility result for the L–P route would itself be worth writing up.

References

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